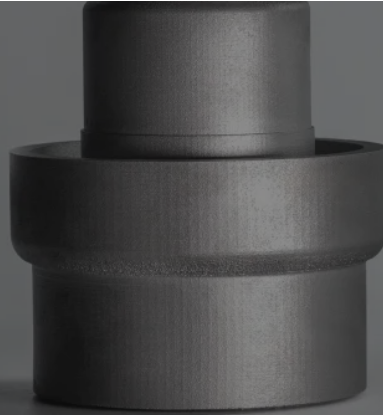


METAL SOLUTIONS

EOS Steel 42CrMo4

Material Data Sheet



EOS STEEL 42CRMO4

EOS Steel 42CrMo4 is a martensitic carbon hardening quench and tempering steel with high strength and toughness. EOS 42CrMo4 corresponds to EN 10083-3 (1.7225, AISI 4140). 42CrMo4 can be heat treated for use in components with different requirements.

MAIN CHARACTERISTICS

- High strength
- High toughness

TYPICAL APPLICATIONS

- Engineering components
- Shafts
- Gears

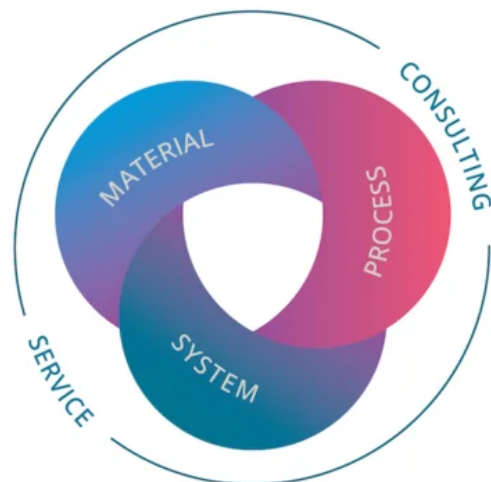
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

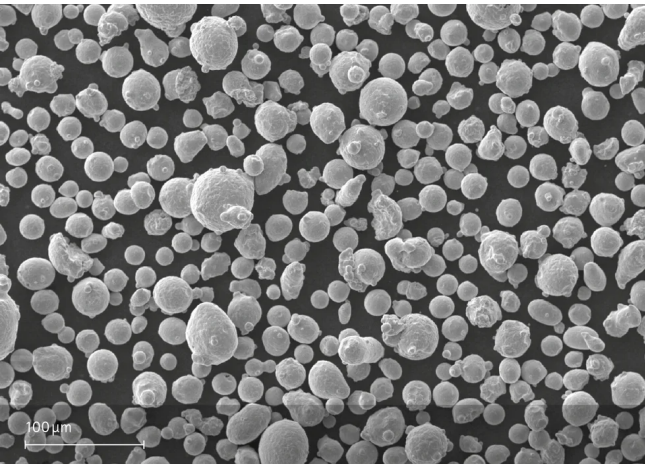
The chemical composition of EOS Steel 42CrMo4 is in compliance with EN 10083-3.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Fe		Balance
Cr	0.9	1.2
Mn	0.6	0.9
C	0.38	0.45
Si	-	0.4
Mo	0.15	0.3
S	-	0.035
P	-	0.025

Powder Particle Size

Generic Particle Size Distribution	15 - 53 µm
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SEM image of powder

HEAT TREATMENT

Description

EOS 42CrMo4 can be heat treated to match various needs of different applications. The two step heat treatment can be performed under vacuum or inert gas atmosphere. First step is austenizing and quenching to achieve martensitic structure. Second step is tempering. Optionally 42CrMo4 can be stress relieved before quench and tempering.

Steps

Optional Stress Relieve

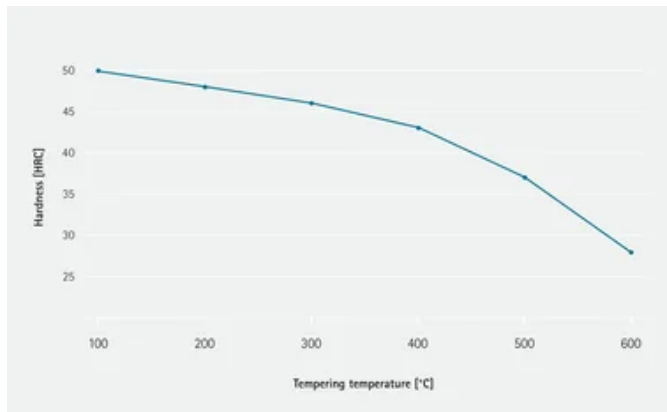
2 h at 600 °C followed by cooling in air

Quench

0.5-1 h at 860 °C (± 20 °C) measured from the part followed by quenching in oil or water to room temperature (below 32 °C).

Tempering

1-2 h at 550 °C (± 20 °C) followed by cooling in air. This tempering gives optimum combination of strength and toughness. Tensile data in this MDS is from this heat treatment. If other hardness is required choose tempering temperature from the graph below.



EOS Steel 42CrMo4 for EOS M 290 I 40/80 µm

EOS M 290 - 40 µm - TRL 3

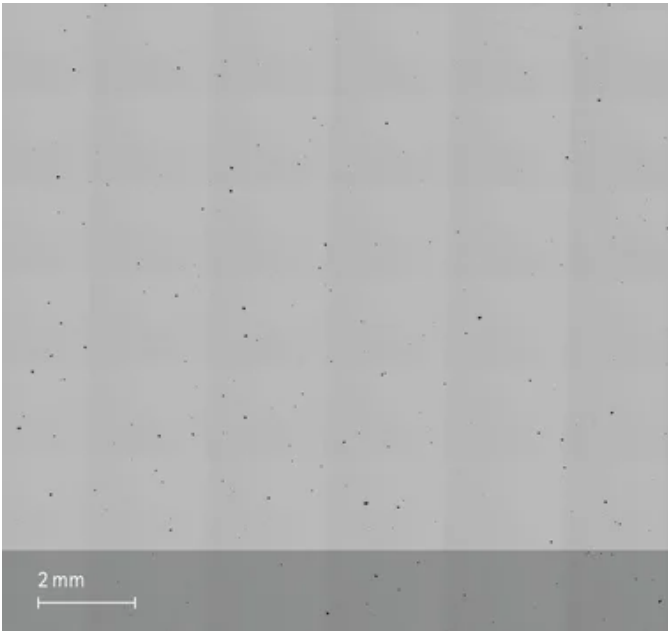
Process Information Metal

This process product is optimized for flexible and fast production of 42CrMo4 parts with EOS M 290. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm SkinCore. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

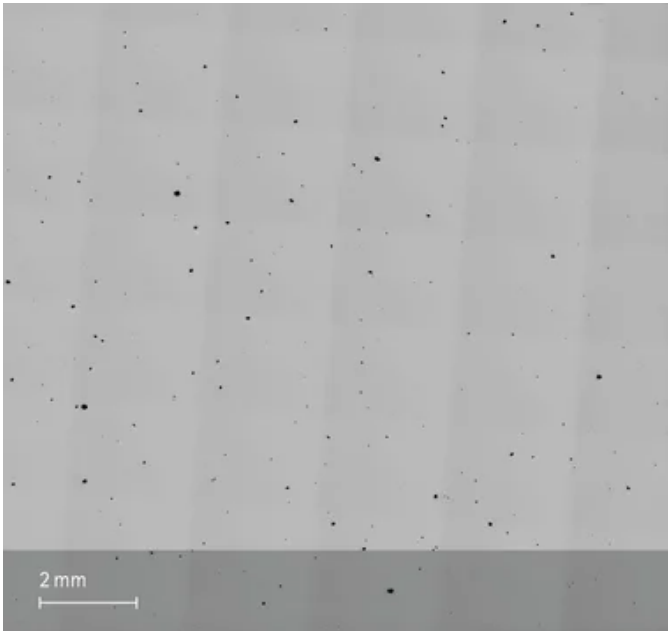
System Setup	EOS M 290
EOS Material set	42CrMo4_040_080_CoreM291
Required Software Plan	EOS BUILD
Software Requirements	EOSPRINT 2.13 or newer EOSYSTEM 2.17 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Nitrogen
Sieve	90 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.2 - 7.68 mm³/s
Typical Dimensional Change after HT [%]	+ 0.2 % %

Chemical and Physical Properties of Parts



Micrograph of polished surface, 40 μm as manufactured



Micrograph of polished surface, 80 μm as manufactured

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 μm	< 0.1 % %	-
Average Defect Percentage	80 μm	< 0.2 % %	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1185	1245	12	-	-	-
Horizontal	1180	1240	12	-	-	-

Tempered at 550 °C

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1170	1230	11	-	-	-
Horizontal	1170	1230	11	-	-	-

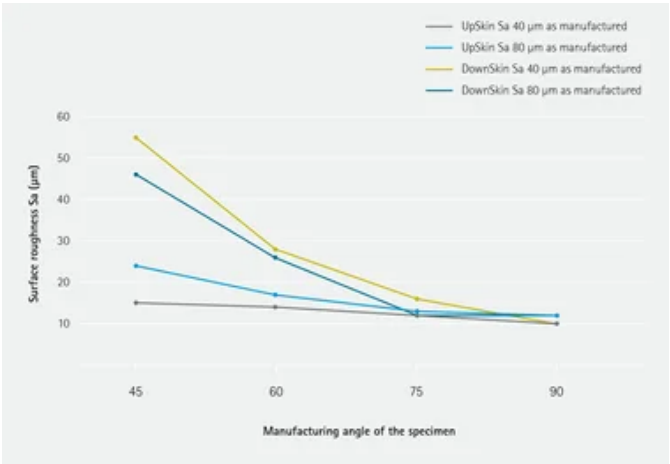
Tempered at 550 °C

Impact Toughness

EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	Vertical: 56 Horizontal: 48
Layer Thickness	40 µm

EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	Vertical: 47 Horizontal: 58
Layer Thickness	80 µm

Surface Roughness



HEADQUARTERS

EOS GmbH
Electro Optical Systems

Robert-Stirling-Ring 1
82152 Krailling / Munich
Germany

Tel.: +49 89 893 36-0
Email: info@eos.info
URL: www.eos.info

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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